

AI Study Companion — Final Submission Package

Complete PRD Deliverables, 13-Step Demo Walkthrough & Architecture Specification

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This document provides all 9 required deliverables specified in the Final Submission Requirements.

1. Working Application

- **Frontend SPA:** `http://localhost:3000`
 - **Backend API & Health:** `http://localhost:4000` (`http://localhost:4000/health`)
 - **Public Cloud Deployment Guide:**
 - **Frontend:** One-click deployment to **Vercel** or **Netlify** from the repository root / `client/` directory.
 - **Backend:** Deploy to **Render**, **Railway**, or **Fly.io** using Node.js 18+ runtime (`npm start` in `server/`).
 - **Environment Variables Required:** `PORT=4000` , `AI_PROVIDER=gemini` , `GEMINI_API_KEY=<key>` , `GEMINI_MODEL=gemini-3.8-flash` , `GEMINI_EXECUTION_MODEL=gemini-3.1-pro` .
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2. Demo Video Walkthrough Guide & Storyboard

The demo video follows the exact required 13-step sequence demonstrating the complete, unbroken learning loop:

Step	Action	Feature Shown in UI	What to Highlight
1	Create Space	User Home → + New Space	Workspace customization (Color, Icon, Title: "AI Foundations").
2	Create Project	Space View → + New Project	Explicit learning goal & target completion date.
3	Upload Material	Project Workspace → Materials Tab	Drag & drop course notes: <i>Machine Learning & Neural Architecture Notes.pdf</i> .
4	Process Material	Background Pipeline Progress	Asynchronous 5-stage progress (Queued → OCR → Structure → Knowledge → Ready).
5	Ask Tutor	Tutor Tab (Grounded Chat)	Ask: "What is a residual connection and why is it used?".
6	Grounded Answer + Citation	Assistant Response Pane	Verifiable citation badge: Source: Machine Learning Notes.pdf – Page 14 .
7	Unsupported Question	Tutor Chat Out-of-Scope Query	Ask: "How do I bake a chocolate cake?" → Zero-hallucination refusal explaining insufficient evidence.
8	Adaptive Quiz	Quiz Tab → Generate Quiz	Dynamic difficulty selection matching concept mastery.
9	Open-Ended Assessment	Quiz Attempt Submission	5-point qualitative AI rubric feedback (Understanding, Accuracy, Relevance, Concepts, Reasoning).
10	Mastery & Growth	Growth Tab	Quantitative mastery bar (88%) & trajectory tags (improving , stable , needs_attention).
11	Analytics	Analytics Tab	Real-time event log, idempotency deduplication demo, and concept mastery trends.
12	Recommendation	Dashboard / Recommendations	Targeted remediation card answering "What should I do next?" anchored to Page 14.
13	Admin Dashboard	Admin Hub Navigation	Platform operational overview, user journey inspection, activity filters, AI observability, and security console.

3. Public GitHub Repository Structure

The repository is fully self-contained, secret-isolated, and production-ready:

```
ai-study-companion/
├── client/                # React 18 + Vite SPA
│   ├── src/pages/        # UserHome, ProjectWorkspace, AdminDashboard, LoginPage, SignupPage
│   └── src/pages/workspace/ # SpaceDashboardView, ProjectDashboardView, MaterialsView, TutorView, QuizView,
│                               GrowthView, AnalyticsView, InnovationsView
│   └── package.json
├── server/               # Express.js REST & SSE Backend
│   ├── middleware/       # authMiddleware (403), rateLimiter (429)
│   ├── services/         # aiProvider, tutorService, quizEngine, masteryService,
│   │                     # retrievalEngine, documentProcessor, contextComposer,
│   │                     # backgroundQueue, workflowEngine, learningEventBus,
│   │                     # evaluationSuite, securityGuard, cacheService
│   ├── data/             # db.json (Atomic readable JSON database with .bak recovery)
│   ├── test.js           # 46 Core Business Logic, Security & PRD Tests (100% Passing)
│   ├── test_security_reliability.js # End-to-end HTTP Security & Perf Tests (100% Passing)
│   └── index.js
├── .env.example          # Secret configuration template
├── .gitignore            # Strict credential & secret isolation
├── ARCHITECTURE.md       # System architecture & technology justification
├── ENGINEERING_DECISIONS.md # Deep dive into trade-offs, decisions & roadmap
├── PROMPTS.md            # Comprehensive log of development prompts
└── README.md             # Setup, execution & feature overview
```

4. Architecture Documentation

Complete architectural diagrams and module justifications are documented in detail in [ARCHITECTURE.md](#) and [ENGINEERING DECISIONS.md](#).

High-Level Separation of Concerns:

- **Presentation:** React 18 + Vite SPA with responsive luxury glassmorphism design tokens.
- **Application/API:** Express router with rate limiting (`429`), tenant isolation middleware (`403`), and Server-Sent Events (SSE) streaming.
- **Business Logic:** Decoupled domain services for learning context, adaptive assessment, mastery tracking, and background workflows.
- **Data & Storage:** Atomic JSON store with temp-file rename + automated `.bak` backup recovery, alongside project-scoped keyword and cosine vector search.
- **Observability & Security:** Granular telemetry in `ai_logs` , 4-pillar benchmark suite in `evaluationSuite.js` , 6 root-cause diagnostics, and prompt injection defense in `securityGuard.js` .

5. AI Usage Documentation

A. AI Used to Build the Product (Development Phase)

- **AI-Assisted Engineering Workflow (Copilot / Assistant Tools):**
- Guided architectural scaffolding, schema design validation, and PRD requirement mapping.
- Test-driven generation of backend domain services (`quizEngine` , `masteryService` , `retrievalEngine` , `workflowEngine`).
- Frontend component scaffolding, responsive layout optimization, and Lucide icon integration.
- Test suite development and cross-layer integration testing.

B. AI Used by the Final Product (Runtime Production Phase)

- **Tiered Cloud Generative Models (`aiProvider.js`):**
- `gemini-3.1-pro` : High-reasoning tasks including 5-point qualitative rubric assessment, open-ended question evaluation, and 4-pillar benchmark suite.
- `gemini-3.8-flash` : Ultra-low-latency conversational RAG Tutor responses and real-time Server-Sent Events (SSE) token streaming.
- **High-Fidelity Resilient Neural Simulator:**
- Local neural fallback engine ensuring 100% operational uptime when third-party cloud APIs experience network disconnects, rate limits, or 8000ms timeouts.
- **AI-Specific Security Guard (`securityGuard.js`):**
- Prompt injection detection, XML boundary isolation (`<untrusted_user_query>`), and adversarial jailbreak neutralization.

6. Development Prompts

Categorized prompts used during architecture, frontend, backend, database, AI, debugging, and testing are documented in [PROMPTS.md](#).

7. Evaluation Approach

The platform treats AI evaluation as a continuous engineering discipline rather than ad-hoc inspection:

1. **4-Pillar Evaluation Suite** (`evaluationSuite.js`):
 2. **Tutor**: Groundedness, citation accuracy (verifying *Page 14*), and out-of-scope question refusal.
 3. **Retrieval**: Relevance scoring (≥ 0.25 threshold) and source metadata integrity.
 4. **Assessment**: 5-point rubric grading quality and strict JSON schema reliability.
 5. **Recommendations**: Relevance, actionability, and alignment with identified weak concepts.
 6. **Regression Detection**:
 7. Automates pre-deployment benchmark runs (`POST /api/admin/ai-eval`) comparing current benchmark pass rates against baseline (`regressionDetected: false`).
 8. **6 Root-Cause Diagnostics**:
 9. Live programmatic endpoints (`GET /api/admin/diagnostics`) answering why responses were slow, which model was used, retrieval quality causes, workflow failure reasons, token costs, and document processing error root causes.
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8. Known Limitations

1. **In-Memory Queue Storage**: Jobs are managed in-process via EventEmitter rather than distributed Redis; restart clears queued non-persisted jobs (persisted jobs remain recorded in `learning_events`).
 2. **Local File Storage**: Uploaded PDFs are stored in the local `uploads/` folder rather than an S3 bucket with CDN edge distribution.
 3. **Keyword-Dense Vector Density**: Retrieval utilizes TF-IDF and keyword-coverage cosine scoring, which is deterministic and lightweight but lacks dense multilingual embedding nuances of external vector DBs.
 4. **Browser Subagent CDN Limitation**: Headless browser automation via Playwright may encounter CDN driver fetch blocks in restricted offline environments.
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9. Future Improvements (Roadmap)

1. **Distributed Queue**: Migrate from in-memory queue to BullMQ / Redis or AWS SQS with Dead Letter Queues (DLQ).
 2. **Dense Vector Search**: Incorporate Qdrant or pgvector with hybrid BM25 + dense neural reranking.
 3. **Multi-Modal Document Understanding**: Native OCR extraction of embedded diagrams, math equations (LaTeX), and tables from scanned textbook PDFs.
 4. **Voice & Audio Learning Companion**: Native WebRTC audio stream for real-time conversational voice tutoring.
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10. Creativity & Differentiation (Cognitive Science & Visual Studio)

Rather than simply multiplying features, differentiation stems from solving fundamental cognitive problems in AI-assisted learning:

1. **The "Illusion of Competence" Problem & The Feynman Technique Studio**:
2. *Problem*: Passive reading creates fluency bias; students believe they understand a concept until forced to explain it without jargon.
3. *Solution*: An inverted active-recall mode where the AI acts as an inquisitive beginner learner ("Elena") asking the user to explain concepts like *Residual Connections* or *Scaled Attention*.
4. *AI Evaluation*: Evaluates the learner's explanation for **Jargon Simplicity** (penalizes raw buzzword dumping), **Everyday Analogies** (rewards intuitive metaphors like highways or toll booths), and **Conceptual Blindspots**.
5. **The "Abstract Geometry" Problem & Interactive Neural Matrix Sandbox**:
6. *Problem*: Transformer mathematics (Query-Key-Value dot products, $\mathcal{O}(N^2)$ scaling, softmax variance) are hard to internalize through pure text.

7. *Solution*: A live visual sandbox with sliders for Sequence Length (N), Model Dimension (d_{model}), Attention Heads (h), and Temperature (τ), rendering real-time $N \times N$ attention heatmaps, memory footprints, and compute FLOPs.
8. **The "Forgetting Curve" Problem & Ebbinghaus Spaced Repetition Forecaster:**
9. *Problem*: Static flashcards are disconnected from learner assessment evidence and source course notes.
10. *Solution*: Dynamic memory half-life modeling ($S = S_0 \cdot e^{-t / \tau}$) calculating projected retention over 14 days, alerting the learner when a concept is within 3 days of dropping below 60% retention.
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11. Security Architecture & Hardening Compliance (PRD Section 15)

The application fully satisfies the **Security Architecture & Hardening Guide**:

- **Authentication**: Salted cryptographic password hashing (`crypto.scryptSync` with timing-safe verification) + Bearer token auth.
- **Authorization & Data Isolation**: Strict multi-tenant isolation (`requireProjectAccess` / `requireSpaceAccess`), returning HTTP 403 Forbidden with security audit logging on foreign access attempts.
- **Prompt Injection Defense**: SecurityGuard pattern detection neutralizing prompt injections with `[REDACTED_SECURITY_OVERRIDE_ATTEMPT]` , escaping XML tags, and isolating untrusted content within delimited envelopes (`<system_instructions>` , `<untrusted_user_query>` , `<retrieved_evidence_untrusted_data>`).
- **File Upload Security**: 25MB maximum size limits, prohibited executable extensions (`.exe` , `.bat` , `.sh` , `.py` , `.bin`), and randomized UUID/timestamp storage paths.
- **API & Network Hardening**: Production security headers (`X-Frame-Options: DENY` , `X-Content-Type-Options: nosniff` , CSP, Referrer-Policy) and token-bucket rate limiting (150 req/min with HTTP 429 and `Retry-After: 60`).
- **Observability & Security Auditing**: Immutable `security_logs` audit trail recording blocked prompt injections, 403 unauthorized project access, 429 rate limit triggers, and login events with IP/UserAgent telemetry.
- **Automated Verification**: 46 / 46 unit & security tests passing with zero warnings or errors.